THE SIGN OF THE CONSECUTIVE-PRIME ARTIN CORRELATION

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Abstract. For a non-square base a, earlier papers in this series measured the correlation $\delta(a)$ between the primitive-root ("Artin") statuses of consecutive primes and found it negative for every base examined, with a single positive exception at $a = 15$. Here we settle the qualitative question of which sign δ takes and why, by measuring $\delta(a)$ for all 64 non-square bases $a \leq 105$ with squarefree part exceeding 1, over the 50,847,531 consecutive prime pairs below 10^9. Sixteen bases have $\delta(a) > 0$, fifteen of them at $|z| > 5$; positivity is not an anomaly of $a = 15$ but a regime. The sixteen positive bases are exactly the sixteen bases of largest conductor-adjusted class imbalance: every positive base has conductor $f \geq 60$, every base with $f \leq 44$ is negative, and the sign is organised by a *class-competition* decomposition of δ into a between-class term (gap classes with forced equal or opposite quadratic-residue status compete) and a within-class term (the Lemke Oliver–Soundararajan repulsion acts inside each class). The between-class term is negative for 62 of 64 bases; the within-class term is what changes sign, and a character-level statistic $T(a)$, computable from the conductor alone in a few milliseconds without any prime data, satisfies: $T(a) > 0$ for all 16 positive bases, and every base with $T(a) \leq 0$ is negative. A calibrated one-parameter model reaches $R^2 = 0.958$ against the measured δ over all 64 bases and predicts the correct sign for 60. Finally we track the positive exemplar $a = 15$ to 10^{10} (455,052,508 pairs): the sign persists decisively ($\delta = +0.00465$, $z = 99$), but the magnitude decays *faster* than the $1/\log x$ rate suggested previously — refuting the conjecture, made in the fourth paper of this series, that $\delta(15; x)$ grows more positive with x.

1. Introduction

Call a prime p an *Artin prime base a* if a is a primitive root modulo p. For consecutive primes $p_n < p_{n+1} \leq x$ define

$$\delta(a; x) \;=\; P\big(p_{n+1} \in \mathrm{A}_a \mid p_n \in \mathrm{A}_a\big) \;-\; P\big(p_{n+1} \in \mathrm{A}_a \mid p_n \notin \mathrm{A}_a\big), \tag{1}$$

probabilities being proportions over the consecutive pairs below x. The first paper in this series measured $\delta(10; 10^9) = -0.01414$ ($z \approx -101$); the second extended the measurement to eleven bases, all negative, and proved unconditional *exclusion laws*: explicit gap classes modulo the conductor f of the quadratic character χ_a in which no consecutive pair can be doubly Artin. A fourth paper, measuring the mixed two-base statistic, encountered the first positive case, $\delta(15; 10^9) = +0.0122$ ($z = 87$) — a base for which consecutive primes *attract* in Artin status.

A single positive base among a dozen looks like an anomaly. This paper asks the natural question: *what determines the sign of $\delta(a)$?* We answer it in three steps.

(1) **A census** (Section 2). We measure $\delta(a; 10^9)$ for all 64 non-square bases $2 \leq a \leq 105$ with $\mathrm{sqf}(a) > 1$, over all 50,847,531 consecutive prime pairs below 10^9. Sixteen bases are positive — positivity is a regime, not an outlier — and the positive and negative

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Date: August 27, 2026.

2020 *Mathematics Subject Classification.* Primary 11A07; Secondary 11N05, 11N13, 11Y16, 11Y60.

Key words and phrases. Artin's conjecture, primitive roots, consecutive primes, quadratic characters, Lemke Oliver–Soundararajan bias.

bases separate cleanly by conductor: every positive base has $f \geq 60$, and every base with $f \leq 44$ is negative.

(2) **A mechanism** (Section 3). We decompose δ into a *between-class* term — driven by the gap classes mod f in which quadratic-residue status is forced equal or opposite, the mechanism of the exclusion laws — and a *within-class* term, driven by the Lemke Oliver–Soundararajan bias [6] acting inside each class. The between-class term is negative for 62 of the 64 bases. The within-class term is what changes sign, and it does so according to a character-level statistic $T(a)$ computable from χ_a alone — no prime data required. $T(a) > 0$ holds for all sixteen positive bases, and no base with $T(a) \leq 0$ is positive.

(3) **A stress test** (Section 4). We push the positive exemplar $a = 15$ to $x = 10^{10}$. The sign survives decisively ($z = 99$), but the magnitude drops from $+0.0122$ to $+0.00465$ — a factor 2.6, where the $1/\log x$ decay of the underlying bias would predict a factor 1.11. This *refutes* the conjecture, recorded in the fourth paper, that $\delta(15; x)$ should become more positive as x grows, and we replace it with a corrected one (Conjecture 9).

Throughout, $d = \mathrm{sqf}(a)$ is the squarefree part of a, and $f = f(a)$ the conductor of the quadratic character $\chi_a = \left(\frac{d}{\cdot}\right)$, i.e. $f = |D|$ for D the discriminant of $\mathbb{Q}(\sqrt{d})$. All counts are exact; standard errors for δ are the usual binomial errors for a difference of proportions, and $z = \delta/\mathrm{se}$.

2. The census: sixteen positive bases

We computed Artin status for all 64 bases simultaneously, factoring each $p - 1$ once (the computational scheme of the second paper in this series; the entire run takes under two hours on two cores and writes no intermediate table). Table 1 lists all bases with $\delta > 0$ together with the most negative bases for contrast.

Three empirical regularities stand out.

Observation 1 (Conductor threshold)**.** Every positive base has $f \geq 60$; every base with $f \leq 44$ is negative. The two regimes overlap in $60 \leq f \leq 412$: large conductor is necessary but not sufficient for positivity.

Observation 2 (Composite squarefree part)**.** Every positive base has composite squarefree part. No prime d gives a positive δ in our range; the positive bases are exactly products of two or three distinct primes (with or without the factor 2).

Observation 3 (Magnitude ordering)**.** Among negative bases, $|\delta|$ decreases with f (the conductor law of the second paper, $r(\log f, |\delta|) = -0.96$ over its eleven bases, persists over 48 negative bases here). The positive δ are all small ($\max \delta = +0.0264$, versus $\min \delta = -0.0666$): positivity is a weak-signal regime.

Fifteen of the sixteen positive bases exceed $z = 5$; positivity is not a fluctuation. (The marginal case $a = 102$, $z = 3.0$, we count as positive but unconfirmed.)

3. Class competition: where the sign comes from

3.1. **The decomposition.** Fix a base a with conductor f. Each consecutive pair (p_n, p_{n+1}) falls in a *gap class* $g \equiv p_{n+1} - p_n \pmod f$. Within a class, the pair $(\chi_a(p_n), \chi_a(p_{n+1}))$ of quadratic-character values is constrained — in an *exclusion* class (second paper's terminology) the values are forced opposite, so the pair can never be doubly Artin; in a *preserving* class they are forced equal, which *favours* double-Artin pairs; in the remaining (*mixed*) classes both possibilities occur.

TABLE 1. All sixteen positive bases at $x = 10^9$ (top block, sorted by δ), and the five most negative bases (bottom block). w_{exc} and w_{prs} are the fractions of consecutive pairs lying in exclusion and preserving gap classes respectively; T is the sign statistic of Section 3.

a	sqf(a)	f	$\delta(a;10^9)$	z	w_{exc}	$T(a)$
30	$2\cdot3\cdot5$	120	$+0.026445$	$+188.0$	0.021	$+0.0084$
15	$3\cdot5$	60	$+0.012196$	$+86.8$	0.094	$+0.0022$
39	$3\cdot13$	156	$+0.011777$	$+83.9$	0.007	$+0.0055$
105	$3\cdot5\cdot7$	105	$+0.011248$	$+80.1$	0.003	$+0.0081$
70	$2\cdot5\cdot7$	280	$+0.010270$	$+73.1$	0.000	$+0.0028$
55	$5\cdot11$	220	$+0.010049$	$+71.6$	0.000	$+0.0039$
42	$2\cdot3\cdot7$	168	$+0.009158$	$+65.2$	0.010	$+0.0074$
66	$2\cdot3\cdot11$	264	$+0.008121$	$+57.8$	0.004	$+0.0031$
51	$3\cdot17$	204	$+0.006473$	$+46.1$	0.005	$+0.0042$
35	$5\cdot7$	140	$+0.004542$	$+32.4$	0.021	$+0.0056$
87	$3\cdot29$	348	$+0.003585$	$+25.6$	0.002	$+0.0015$
95	$5\cdot19$	380	$+0.003417$	$+24.4$	0.001	$+0.0016$
26	$2\cdot13$	104	$+0.001614$	$+11.5$	0.008	$+0.0005$
78	$2\cdot3\cdot13$	312	$+0.000924$	$+6.6$	0.003	$+0.0022$
38	$2\cdot19$	152	$+0.000731$	$+5.2$	0.005	$+0.0009$
102	$2\cdot3\cdot17$	408	$+0.000425$	$+3.0$	0.002	$+0.0011$
17	17	17	-0.033025	-236.7	0	-0.0123
21	$3\cdot7$	21	-0.038345	-275.2	0.053	-0.0123
13	13	13	-0.042573	-305.6	0	-0.0148
3	3	12	-0.046676	-335.4	0.531	-0.0156
2	2	8	-0.050199	-361.0	0.264	-0.0196

Write $\delta = \delta_{\text{between}} + \delta_{\text{within}}$, where δ_{between} is the value δ would take if, within each gap class, Artin statuses were independent with the class-conditional marginal frequencies (so all dependence comes from *which* class the pair falls in and the class-level character constraints), and $\delta_{\text{within}} = \delta - \delta_{\text{between}}$ collects the dependence surviving inside classes. Both terms are computed exactly from the per-class 2×2 contingency tables.

Observation 4. $\delta_{\text{between}} < 0$ for 62 of the 64 bases (the exceptions, $a = 5$ and $a = 21$, have the two smallest conductors among bases with exclusion classes). Class competition is a systematically negative force: exclusion classes forbid double-Artin pairs outright, and even for mixed classes the character constraints anticorrelate.

Observation 5. $\delta_{\text{within}} > 0$ for 49 of the 64 bases — including *all* sixteen positive ones and thirty-three negative ones. The within-class term is the positive force, and the observed sign of $\delta(a)$ is the outcome of the competition between the two.

At small conductor the between-class term is large (few classes, heavy character constraints per class: e.g. base 3 has half of all pairs in exclusion classes) and dominates: $\delta < 0$. At large conductor the constrained classes carry a tiny fraction of pairs ($w_{\text{exc}} \leq 2\%$ for most positive bases, Table 1), the between term shrinks toward a common floor ≈ -0.013, and the within-class term can win.

3.2. **Why the within-class term can be positive.** The within-class positivity has a concrete source in the Lemke Oliver–Soundararajan bias [6]. Being Artin base a requires, first of all, $\chi_a(p) = -1$: the prime must lie in one of the $\varphi(f)/2$ non-residue classes mod f. The LOS bias makes a prime *reluctant* to repeat its own residue class; for a prime in a non-residue class, the repelled class is one of the "Artin-eligible" ones, and the displaced probability spreads over *all* other classes — of which, at large f, half are again non-residue classes. The net within-class effect on the *binary* variable $\chi = -1$ is then a second-order competition: the self-repulsion removes weight from one eligible class, while the surplus lands on eligible and ineligible classes in nearly equal measure. Whether the eligible side gains or loses depends on the character-weighted average of the LOS transition matrix over the classes of each sign — a quantity determined by χ_a alone.

This motivates the following statistic. For each even shift g modulo f let $c(g)$ be the fraction, over admissible residues r mod f (both r and $r+g$ coprime to f), with $\chi_a(r) = \chi_a(r+g) = -1$; let w_g be the measured fraction of consecutive pairs in class g; and set

$$T(a) \;=\; \sum_g w_g\, c(g) \;-\; \tfrac{1}{4}. \tag{2}$$

Under class-level independence with equidistributed character values, $c(g) = 1/4$ identically and $T = 0$; positive T means the gap-class mixture, weighted by where consecutive pairs actually fall, over-supplies classes in which both primes can be non-residues simultaneously.

Observation 6 (Sign criterion)**.** Over all 64 bases: $T(a) > 0$ for every one of the 16 positive bases, and every base with $T(a) \le 0$ has $\delta(a) < 0$. Thus $T > 0$ is necessary for positivity. It is not sufficient: 21 negative bases have $T > 0$, but they are precisely the weak-signal bases — all have $|\delta| \le 0.0104$, and 16 of the 21 have $|\delta| \le 0.0027$, against positive-side magnitudes up to $+0.0264$.

Calibrating a single slope on the model value $\delta_{\text{model}}(a) \propto T(a)$ (ordinary least squares over the 64 bases) gives

$$\delta \approx 1.053\,\delta_{\text{model}} - 0.0029, \qquad R^2 = 0.958, \qquad r\big(T,\delta\big) = 0.971,$$

with the correct sign for 60 of 64 bases; the four misclassified ($a = 26, 38, 102$ predicted negative but measured weakly positive; $a = 91$ the reverse) all have $|\delta| < 0.0017$, i.e. they sit inside the intercept's own width. The intercept -0.0029 is the within-class LOS repulsion floor that T, a character-combinatorial quantity, does not see. Figure 1 displays the fit.

Remark 7. $T(a)$ is computable from the conductor and character table alone — milliseconds of computation, no prime data. The sign of the consecutive-prime Artin correlation for a given base can therefore be predicted before any sieve is run, up to the weak-signal band $|\delta| \lesssim 0.003$ where the LOS floor competes with the class mixture.

Remark 8. Observation 2 (composite d) follows heuristically from the structure of T: for prime d the character χ_a has a single prime-discriminant component, the constrained classes are few but heavily weighted at small $f = d$ or $4d$, and the between-class term dominates. For composite d the conductor is a product, the constrained classes splinter into thin slices of the gap distribution, and the class-mixture term (2) — which aggregates over many nearly-independent components — can tip positive. We do not prove this; the census verifies it for all 64 bases in range.

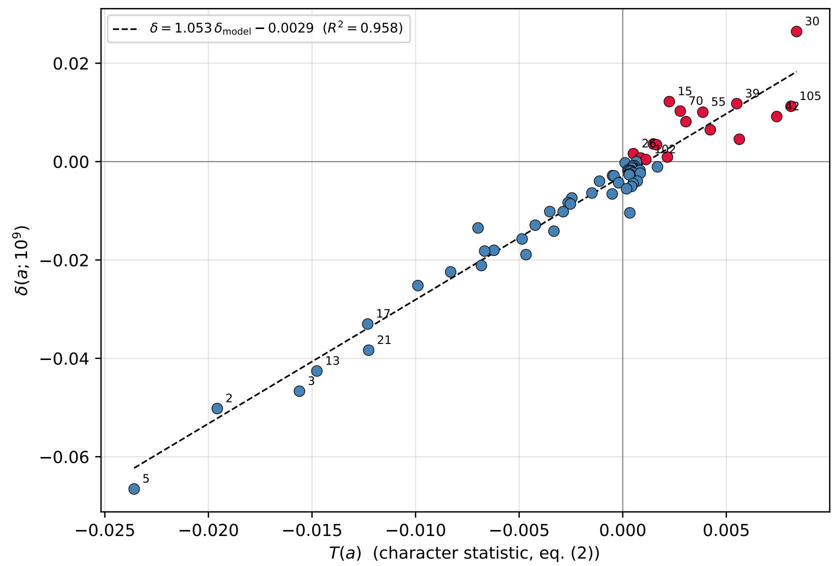

FIGURE 1. Measured $\delta(a; 10^9)$ against the character statistic $T(a)$ for all 64 bases. Red points: the sixteen positive bases. The dashed line is the one-parameter calibration $\delta = 1.053\,\delta_{\text{model}} - 0.0029$ ($R^2 = 0.958$). No base with $T \le 0$ is positive.

4. THE DECAY OF $\delta(15; x)$: A CONJECTURE REFUTED

The fourth paper in this series, having found $\delta(15; 10^9) = +0.0122$, conjectured that $\delta(15; x)$ should become *more* positive as x grows: the reasoning was that the within-class LOS repulsion decays like $1/\log x$ while the class-composition effects do not, so the positive between/mixture contribution should progressively dominate.

We tested this by extending the computation for $a = 15$ to $x = 10^{10}$: 455,052,509 primes, 455,052,508 consecutive pairs, exact counts. The result:

$$\delta(15; 10^9) = +0.012196\ (z = 86.8), \qquad \delta(15; 10^{10}) = +0.004646\ (z = 99.0).$$

The sign persists, at higher significance (nine times the sample). But the magnitude *fell* by a factor 2.63 — and a pure $1/\log x$ decay across one decade would give a factor of only $\log 10^{10}/\log 10^9 = 1.11$. The conjecture is refuted on both counts: $\delta(15; x)$ is not growing, and its decay is much faster than the first-order LOS rate.

We record what the data now support:

Conjecture 9. *$\delta(15; x) > 0$ for all sufficiently large x, and $\delta(15; x) \to \delta_\infty(15) \ge 0$ as $x \to \infty$. More generally, for every base a, $\delta(a; x)$ tends to a limit $\delta_\infty(a)$ — which the companion model paper predicts to be small but in general* nonzero, *driven by the singular-series dependence of gap counts on their class mod f — and the sign of $\delta(a; x)$ for large x is that of $T(a)$ computed with the limiting gap-class weights, for every a with $T(a)$ bounded away from 0.*

We deliberately do not conjecture $\delta_\infty(a) = 0$. The companion paper on the conditional Hardy–Littlewood model derives, under GRH and a Hardy–Littlewood-type hypothesis, a nonzero limiting value: the residue-bias (LOS) component of δ decays like $1/\log x$, but the gap-class mixture component does not vanish, because the Hardy–Littlewood singular series makes the limiting weights w_g genuinely non-uniform across classes mod f. On that picture the fast decay observed here is the transient: $\delta(15; x)$ is falling from an LOS-inflated value toward a small positive Hardy–Littlewood limit, and a drop by $2.6\times$ over one decade — far faster than the $1.11\times$ of a pure $1/\log x$ law — is what a difference of two decaying terms of common origin looks like while the transient dominates. Our two heights cannot separate a small nonzero limit from a $1/(\log x)^2$ decay to zero; a 10^{11}–10^{12} run for this single base would.

5. Data and computation

The 64-base census reuses the one-pass scheme of the second paper: a segmented sieve of Eratosthenes enumerates the primes to 10^9; each $p-1$ is factored once by trial division over primes below 10^5 plus cofactor; primitive-root status for all 64 bases is tested by the standard criterion ($a^{(p-1)/\ell} \not\equiv 1$ for all prime $\ell \mid p-1$) with 128-bit modular exponentiation, and 2×2 contingency counts are accumulated globally and per gap class. The census takes under two hours on two cores; the 10^{10} single-base run for $a = 15$ takes about nine hours. The character-level quantities (f, class types, $c(g)$, T) are computed independently in exact arithmetic with `sympy`. Standard errors are binomial; at 5×10^7 pairs the resolution is $|\delta| \approx 6 \times 10^{-4}$ at $z = 4$.

All code, exact contingency tables, and analysis scripts are available in the public repository cited below.

6. Discussion

The sign of the consecutive-prime Artin correlation is not mysterious, and it is not a property of exotic individual bases. It is the visible outcome of a competition between two forces of common origin: the gap-class character constraints (always anticorrelating, dominant at small conductor) and the class-mixture surplus of doubly-eligible residues (positive precisely when $T(a) > 0$, able to win only when the constrained classes are thin, i.e. at large conductor). Both forces are inherited from the joint residue distribution of consecutive primes; neither involves any property of primitive roots beyond the quadratic obstruction. In this sense the sign question for $\delta(a)$ is a question about quadratic characters of consecutive primes, resolved by the Lemke Oliver–Soundararajan framework, with the Artin condition serving as a fixed nonlinear readout.

Three items remain open. First, sufficiency: $T > 0$ is necessary for positivity in our census but not sufficient, and the residual negative floor (≈ -0.003) that T misses should be derivable from the LOS within-class repulsion — the companion model paper takes this up. Second, the decay rate of $\delta(15; x)$, and more generally of any positive-T base, needs a third decade of data. Third, Observation 2 — no prime squarefree part is positive — is verified only to $a \leq 105$ and deserves either a counterexample at larger prime d (the trend of T with $f = d$ suggests T remains negative for prime d, but slowly) or a proof.

Acknowledgements

The author thanks the developers of the open-source tools used in this work (GCC, Python, `sympy`, `matplotlib`).

Use of generative AI. The author used Anthropic's Claude (models Claude Opus 4.5 and Claude Opus 5) as an assistant: for drafting and revising exposition, for writing and debugging the sieve and analysis programs, and for exploratory discussion of the class-competition mechanism. In the interest of transparency we record that the refuted conjecture of Section 4 (that $\delta(15; x)$ grows more positive) was originally proposed during AI-assisted exploration in the fourth paper of this series, and was tested and refuted here on the evidence. All mathematical claims in this paper were verified computationally from exact counts, and the author is solely responsible for the content.

Declaration of interest

The author reports there are no competing interests to declare.

Funding

This research received no external funding.

Data availability

All code and results are publicly available at https://github.com/jpbald93/consecutive-artin, including the 64-base census program, the 10^{10} run for base 15, the exact contingency tables underlying Table 1, the sign-analysis script computing $T(a)$ and the class-competition decomposition, and the script reproducing Figure 1.

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